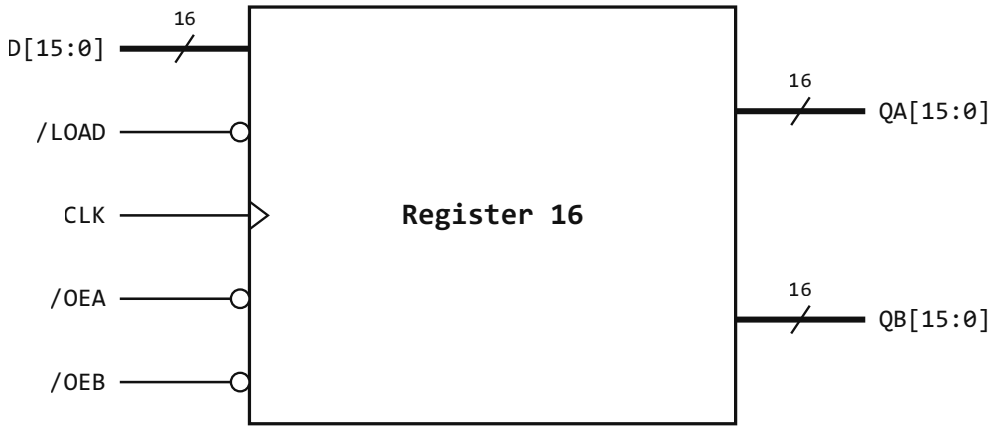
R16 Register 16

16-bit register with two independent tri-state read ports. On the rising edge of CLK it loads D[15:0] when /LOAD is low and holds otherwise. QA drives while /OEA is low, QB while /OEB is low; outputs are independent of the clock.

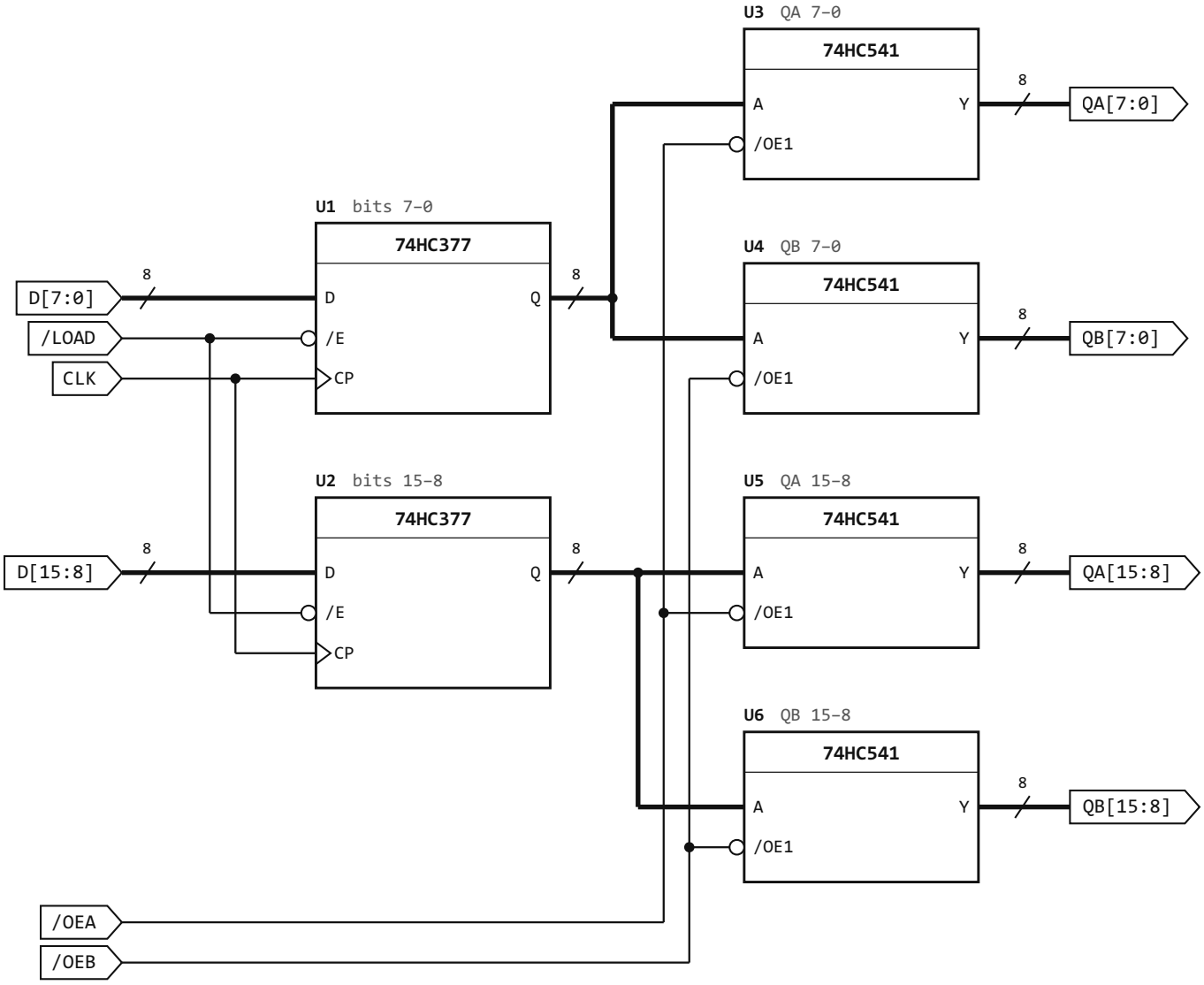
Width	16 bits	**Logic family**	74HC, 5 V	**Clocking**	Rising edge, synchronous load
Read ports	2 × tri-state	**Reset**	None (random at power-up)	**Chips**	6

1 PINOUT



Pin	Dir	Width	Active	Function
D[15:0]	In	16	High	Data to load
/LOAD	In	1	Low	Load enable, sampled on CLK ↑
CLK	In	1	Rising ↑	Clock, rising edge
/OEA	In	1	Low	Enable read port A
/OEB	In	1	Low	Enable read port B
QA[15:0]	Out	16	High	Read port A, tri-state
QB[15:0]	Out	16	High	Read port B, tri-state

2 SCHEMATIC


U3 QA 7–0
74HC541
A
Y
/OE1
8
QA[7:0]
U1 bits 7–0
74HC377
D
Q
/E
CP
8
D[7:0]
/LOAD
CLK
U4 QB 7–0
74HC541
A
Y
/OE1
8
QB[7:0]
U2 bits 15–8
74HC377
D
Q
/E
CP
8
D[15:8]
U5 QA 15–8
74HC541
A
Y
/OE1
8
QA[15:8]
U6 QB 15–8
74HC541
A
Y
/OE1
8
QB[15:8]
/OEA
/OEB

3 TRANSITION TABLES

Load

CLK	/LOAD	D	Q next
↑	0	data	data
↑	1	X	Q_o
0 / 1 / ↓	X	X	Q_o

↑ rising edge · X don't care · Q_o previous value

Read ports

/OEA	QA[15:0]	/OEB	QB[15:0]
0	Q	0	Q
1	Z	1	Z

independent of CLK · Z high-impedance

4 BILL OF MATERIALS

Ref	Qty	Part	Description	Package
U1 U2	2	74HC377	Octal D flip-flop with enable	DIP-20
U3–U6	4	74HC541	Octal tri-state buffer	DIP-20
C1–C6	6	100 nF	Ceramic decoupling, one per chip	2.54 mm
Total	**12**			